Md Ahsanul Haque, Ph.D.
MAT-120

Midterm

Date: November 5, 2025

Name

Student ID

Section

DISTRIBUTION OF MARKS		
Question No.	**Full Marks**	**Obtained Marks**
1		
2		
3		
4		
5		
6		
7		
8		
Total		

Instructions:

- Write your full name (first **and** last name) **inside** the provided box.
- Clearly write your student ID and section number.
- This assignment contains 6 problems. It is **YOUR responsibility** to check that your copy has all 6 problems.
- Make sure to read the instructions for each problem carefully.
- Answer **5** questions. If you answer 6, the first 5 will be graded.
- **Show your work** using the space provided.
- If you need more space, you may use the sheet labeled "Extra Space" at the end of the exam. You **must** label what problem you are continuing the work for (as well as label in the problem itself that you will finish it on the extra page).
- Use correct mathematical notation.
- Write cohesively and follow a logical sequence of steps.
- Check that your answer can be read and understood by the grader.

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1. (10 points) Consider the function:

$$f(x) = \begin{cases} x^2 + 2 & \text{if } x < 1 \\ 3x & \text{if } x = 1 \\ \dfrac{x^2 - 1}{x - 1} & \text{if } x > 1 \end{cases}$$

   (a) Find $\lim_{x \to 1^-} f(x)$ and $\lim_{x \to 1^+} f(x)$

   (b) Does $\lim_{x \to 1} f(x)$ exist? Justify your answer.

   (c) Find $\lim_{x \to \infty} f(x)$ and $\lim_{x \to -\infty} f(x)$

   (d) Identify all points of discontinuity and classify them.

Extra Space

2. (10 points)

Let $f(x) = \dfrac{\sqrt{x+4}-2}{x}$.

(a) Find $\lim_{x\to 0} f(x)$ using algebraic manipulation.

(b) Determine if $g(x) = \dfrac{x^2-9}{x-3}$ is continuous at $x = 3$. If not, is the discontinuity removable?

(c) Find the value of k that makes $h(x) = \begin{cases} kx^2 & \text{if } x \leq 2 \\ x+k & \text{if } x > 2 \end{cases}$ continuous everywhere.

(d) Investigate the continuity of $p(x) = \dfrac{\sin x}{x}$ at $x = 0$.

Extra Space

3. (10 points) Analyze the following limits:
   (a) $\lim_{x\to 0} \dfrac{\sin(3x)}{5x}$
   (b) $\lim_{x\to 0} \frac{\sqrt{x+9}-3}{x}$
   (c) $\lim_{x\to 0} \dfrac{1-\cos(2x)}{x^2}$
   (d) $\lim_{x\to \infty} e^{-x}\sin x$

Extra Space

4. (10 points) Let $f(x) = x^3 - 3x^2 + 2$.
   (a) Find the derivative $f'(x)$ using the limit definition.
   (b) Find the equation of the tangent line at $x = 2$.
   (c) Find all points where the tangent line is horizontal.
   (d) Find the average rate of change on $[0, 3]$ and compare with the instantaneous rate at $x = 1.5$.

Extra Space

5. (10 points)

   Differentiate the following functions:

   (a) $y = \sin(2x^3 - 4x)$

   (b) $y = e^{x^2}\cos(3x)$

   (c) $y = \sqrt[3]{\dfrac{1-x}{1+x}}$

   (d) $y = \ln(\sec x + \tan x)$

Extra Space

6. (10 points) Consider the curve defined by $x^3 + y^3 = 6xy$.
   (a) Find $\dfrac{dy}{dx}$ using implicit differentiation.
   (b) Find all points where the tangent line is vertical.
   (c) Find $\dfrac{d^2y}{dx^2}$ in terms of x and y.
   (d) Suppose a spring-mass system is acted upon by an external periodic force. The motion of the spring is governed by the differential equation:

$$y'' + 9y = 20\cos(4t)$$

Assume a particular solution of the form:

$$y = A\cos(4t)$$

Determine the value of A such that y(t) satisfies the equation.

Extra Space